

Indian Institute of Technology Bombay



Numerical Construction of Initial Data in General Theory of Relativity

Project Report

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Abstract

The observation of gravitational wave events marks a rapidly advancing frontier in modern astronomy, where accurate modelling of compact, high-mass systems plays a key role. In this project, we focus primarily on the 3+1 decomposition and the resulting constraint equations for Maxwell's equations and in general relativity for the metric tensor and Einstein's equations. We show how the electromagnetic constraints can be derived directly from the field-strength (Faraday) tensor. We also study the numerical implementation of Maxwell's equations using two different simulation techniques and analyze the behaviour of the constraints, the associated residuals, and the computational complexity of each method.

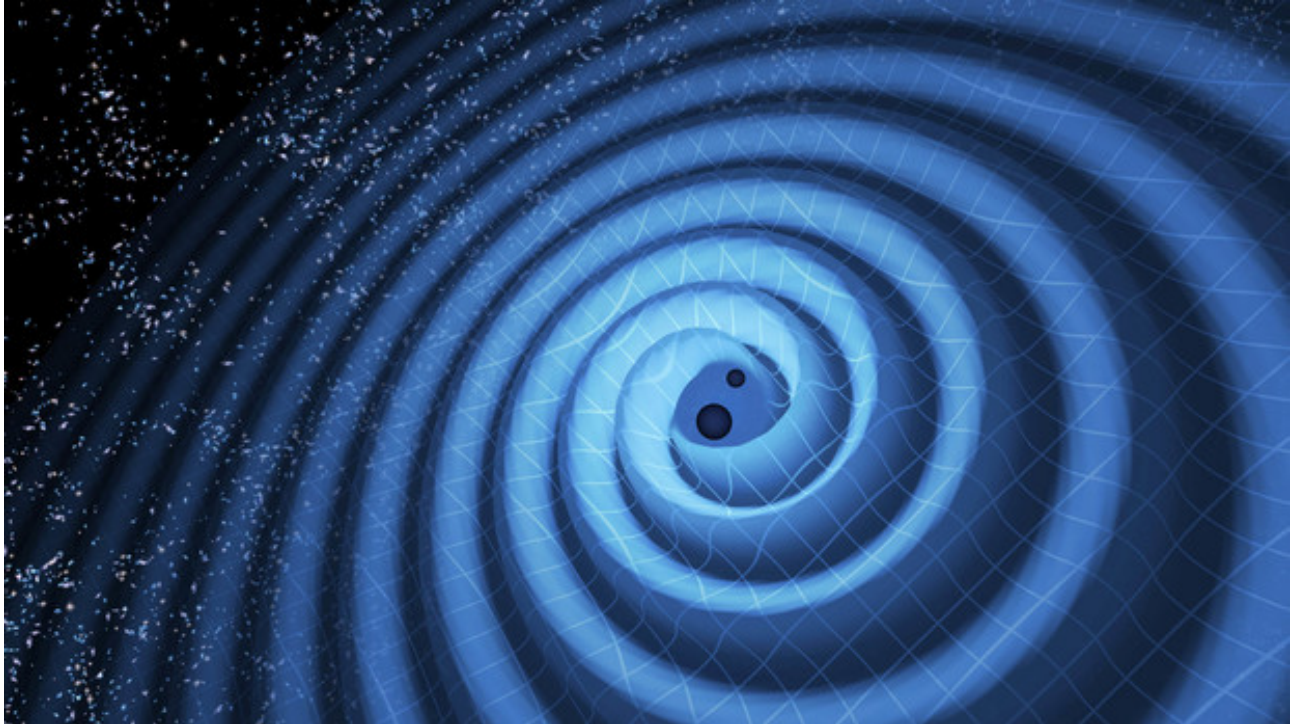


Figure 1: Illustration of a black hole merger producing Gravitational waves, credit: T. Pyle, LIGO

Contents

Abstract	1
1. Project Motivation	3
2. Maxwell Theory and the Faraday Tensor	4
2.1. Conventions for Potentials and the Electric/Magnetic Fields	4
2.2. The 3 + 1 Split: Foliation and Kinematics	5
2.3. Electric and Magnetic Fields as Spatial Projections	5
2.4. Projecting Maxwell's Equations: Constraints and Evolution	5
2.5. Recovering E and B from Potentials in the 3 + 1 Split	6
3. General Relativity in the 3+1 Formalism	7
3.1. Extrinsic curvature and spatial covariant derivative	7
3.2. Projections of the spacetime Riemann tensor	7
3.3. Constraint equations from Einstein's equation	8
3.4. Summary of the Four Fundamental Relations	8
4. Numerical Methods for Maxwell's Equations	9
4.1. Spectral representation and use of the FFT	9
4.2. Initializing via potentials vs directly initializing fields	9
4.3. Computational cost and practical timing	12
4.4. Random initial fields vs updating-from-potentials	13
4.5. Boundary conditions and domain choices	13
4.6. Visualizing Dipole residuals using Potential Initialisation	14
5. References	15

1 Project Motivation

Gravitational waves are ripples in spacetime generated by the accelerated motion of massive bodies. Because they interact very weakly with matter, they can reveal astrophysical processes that are otherwise hidden, yet this same property makes them extremely difficult to detect. By the time they reach Earth, their amplitudes are tiny, and the signals recorded by detectors like LIGO often have very low signal-to-noise ratios.

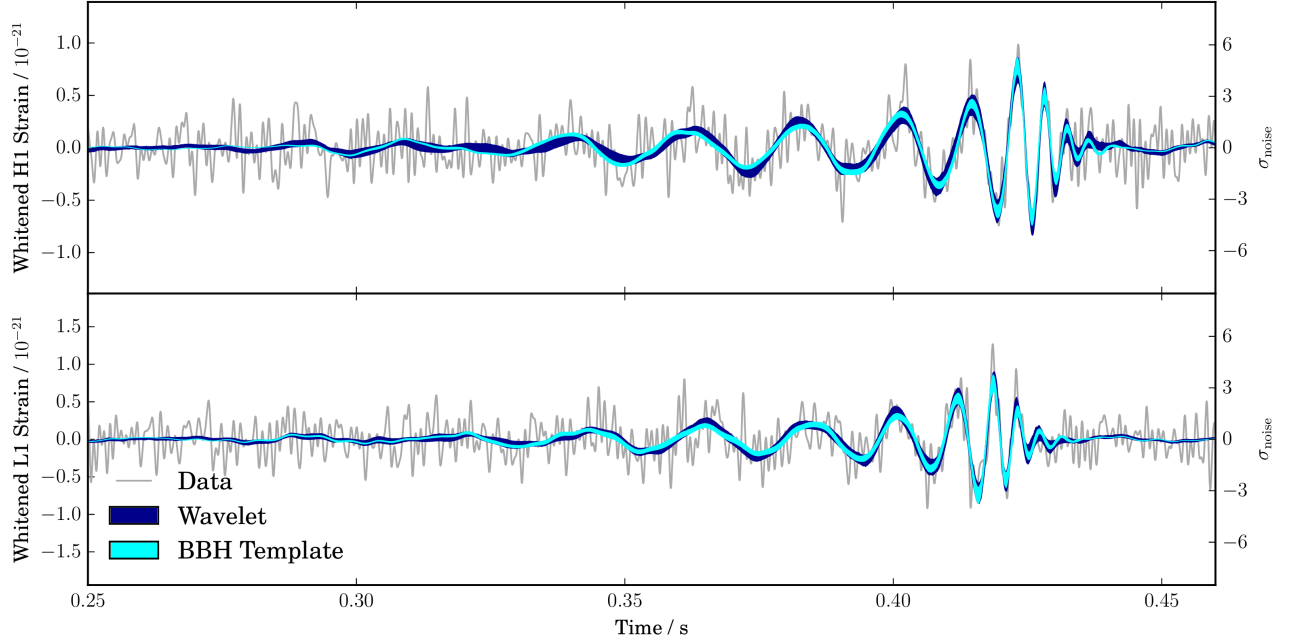


Figure 2: Recovered gravitational waveforms from analysis of GW150914, credit: Christopher Berry

To reliably identify such weak signals, LIGO data analysis relies heavily on Bayesian inference, which compares the observed strain with theoretically predicted waveforms. This creates a strong need for accurate waveform templates describing different gravitational systems. These templates depend on solving Einstein's equations numerically, which in turn requires well-constructed initial data for the systems being simulated.

The aim of this project is therefore to construct the initial data needed to simulate relevant gravitational systems using general relativity. Producing accurate simulations allows us to generate the waveform models essential for Bayesian analysis of low-SNR LIGO data. Without these models, many potential gravitational-wave events could be overlooked, making this step crucial for both detection and interpretation.

2 Maxwell Theory and the Faraday Tensor

We adopt the metric signature $(-, +, +, +)$. The electromagnetic field is described by the antisymmetric Faraday tensor $F_{\mu\nu}$ and the four-current J^μ . The field tensor is defined in terms of the four-potential A_μ by

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu. \quad (1)$$

Maxwell's equations in covariant form are

$$\nabla_\mu F^{\mu\nu} = \mu_0 J^\nu, \quad \nabla_{[\alpha} F_{\beta\gamma]} = 0, \quad (2)$$

where the second (homogeneous) equation is equivalent to the existence of the potential (1) and encodes Faraday's law and the absence of magnetic monopoles. The Hodge-dual of $F^{\mu\nu}$ is defined by

$$*F^{\mu\nu} = \frac{1}{2} \epsilon^{\mu\nu\rho\sigma} F_{\rho\sigma}, \quad (3)$$

and the homogeneous Maxwell equation can be compactly written as $\nabla_\mu *F^{\mu\nu} = 0$.

2.1 Conventions for Potentials and the Electric/Magnetic Fields

To connect with the usual electric and magnetic fields we choose the following decomposition of the four-potential (consistent with the metric signature above):

$$A_\mu = (-\Phi, ; A_i), \quad (4)$$

so that $A_0 = -\Phi$ and A_i is the spatial vector potential. With this sign convention one recovers the familiar relations (in flat space, Cartesian coordinates)

$$E_i = -\partial_t A_i - \partial_i \Phi, \quad B^i = \frac{1}{2} \varepsilon^{ijk} (\partial_j A_k - \partial_k A_j), \quad (5)$$

where ε^{ijk} is the spatial Levi-Civita symbol (with $\varepsilon^{123} = +1$). These follow directly from

$$E_i = F_{i0}, \quad B^i = \frac{1}{2} \varepsilon^{ijk} F_{jk}, \quad (6)$$

combined with (1).

Two useful gauge choices are:

- **Lorenz gauge:** $\nabla_\mu A^\mu = 0$. In this gauge the inhomogeneous Maxwell equations become simple wave equations

$$\square A^\mu = \mu_0 J^\mu. \quad (7)$$

This gauge is manifestly Lorentz-covariant and convenient for deriving retarded solutions such as the Liénard–Wiechert potentials for a moving point charge.

- **Coulomb gauge (radiation gauge):** $\nabla \cdot \mathbf{A} = 0$. In flat space this separates the instantaneous (constraint) equation for the scalar potential from the radiative vector degrees of freedom; it is often convenient for canonical and Hamiltonian treatments.

As an example, the Liénard–Wiechert potentials (in Lorenz gauge) for a point charge q with worldline $z^\mu(\tau)$ can be compactly written as retarded integrals; schematically

$$A^\mu(x) = \frac{\mu_0 q}{4\pi} \frac{u^\mu}{(u \cdot (x - z))} \Big|_{\text{ret}}, \quad (8)$$

where u^μ is the source four-velocity and the right-hand side is evaluated at the retarded emission event.

2.2 The 3 + 1 Split: Foliation and Kinematics

We foliate spacetime by spacelike hypersurfaces Σ_t labeled by a time coordinate t . Let the unit future-directed normal to the slices be denoted by n^μ (so $n_\mu n^\mu = -1$). The spacetime metric may be written in ADM form:

$$ds^2 = -(\alpha^2 - \beta_i \beta^i) dt^2 + 2\beta_i dt, dx^i + \gamma_{ij}, dx^i, dx^j, \quad (9)$$

where α is the lapse, β^i the shift, and γ_{ij} the induced metric on each slice Σ_t . The projector into the slice is

$$\gamma^\mu * \nu = \delta^\mu * \nu + n^\mu n_\nu, \quad (10)$$

which satisfies $\gamma^\mu * \nu n^\nu = 0$ and $\gamma^\mu * \rho \gamma^\rho * \nu = \gamma^\mu * \nu$. In coordinates adapted to the foliation (so that t labels slices) the unit normal has components

$$n_\mu = (-\alpha, 0, 0, 0), \quad n^\mu = \left(\frac{1}{\alpha}, -\frac{\beta^i}{\alpha} \right). \quad (11)$$

2.3 Electric and Magnetic Fields as Spatial Projections

The electromagnetic fields measured by an Eulerian observer (whose four-velocity is n^μ) are the spatial vector fields

$$E^\mu = F^{\mu\nu} n_\nu, \quad B^\mu = {}^* F^{\mu\nu} n_\nu = \frac{1}{2} \epsilon^{\mu\nu\rho\sigma} n_\nu F_{\rho\sigma}, \quad (12)$$

which satisfy $E^\mu n_\mu = B^\mu n_\mu = 0$ and are therefore tangent to Σ_t . We will denote their purely spatial components by E^i and B^i (raising/lowering with γ_{ij} when appropriate). Using the projector one can express the field tensor entirely in terms of E^i and B^i :

$$F_{\mu\nu} = n_\mu E_\nu - n_\nu E_\mu + \epsilon_{\mu\nu\rho} B^\rho, \quad (13)$$

where $\epsilon_{\mu\nu\rho} = \epsilon_{\alpha\mu\nu\rho} n^\alpha$ is the spatial volume form on Σ_t (so the purely spatial components give $F_{ij} = \epsilon_{ijk} B^k$).

2.4 Projecting Maxwell's Equations: Constraints and Evolution

Decompose the four-current relative to the Eulerian observers as

$$\rho = -n_\mu J^\mu, \quad j^\mu = \gamma^\mu * \nu J^\nu, \quad (14)$$

so that $J^\mu = \rho n^\mu + j^\mu$ and $j^\mu n_\mu = 0$. Project the inhomogeneous Maxwell equation along n_ν and into the slice to obtain the constraint and evolution equations, respectively.

Constraint (Gauss) equations

Contracting with n_ν and using the kinematic identities of the foliation yields

$$D_i E^i = \mu_0 \rho, \quad D_i B^i = 0, \quad (15)$$

where D_i is the covariant derivative compatible with the spatial metric γ_{ij} . These are the familiar Gauss constraint (electric) and the absence of magnetic monopoles.

Evolution equations

Projecting tangent to Σ_t gives the evolution equation for E^i , while projecting the homogeneous equation gives the evolution for B^i . In an index-free geometric form one may write:

$$\mathcal{L}_n E^i = \epsilon^{ijk} D_j(\alpha B_k) - \alpha j^i + \alpha K E^i, \quad \mathcal{L}_n B^i = -\epsilon^{ijk} D_j(\alpha E_k) + \alpha K B^i, \quad (16)$$

where $\mathcal{L}_n$ is the Lie derivative along n^μ , K is the trace of the extrinsic curvature $K * ij$ of the slice, and indices are raised/lowered with γ_{ij} . These equations display the coupling of the electromagnetic fields to the geometry (via α, β^i, K_{ij}) when evolving on a dynamical spacetime.

Flat-space (Minkowski) limit. To make contact with the standard Maxwell evolution used in electrodynamics and numerical electrodynamics, set $\alpha = 1$, $\beta^i = 0$, and $\gamma_{ij} = \delta_{ij}$ (so $K_{ij} = 0$). Then the evolution equations reduce to the well-known first-order system

$$\partial_t E^i = \epsilon^{ijk} \partial_j B_k - j^i, \quad \partial_t B^i = -\epsilon^{ijk} \partial_j E_k, \quad (17)$$

supplemented by the constraints $\partial_i E^i = \rho$ and $\partial_i B^i = 0$. These are the exact component forms one obtains when writing Maxwell's equations in the usual $\mathbf{E}, \mathbf{B}$ language.

2.5 Recovering $\mathbf{E}$ and $\mathbf{B}$ from Potentials in the $3 + 1$ Split

Split the four-potential relative to Eulerian observers as

$$\Phi := -n^\mu A_\mu, \quad A_i := \gamma_i^\mu A_\mu, \quad (18)$$

so that $A_\mu = -\Phi n_\mu + A_i \gamma_i^\mu$. In adapted coordinates (with lapse and shift) the gauge-invariant spatial electric and magnetic fields become

$$E_i = -D_i \Phi - \mathcal{L}_n A_i, \quad B^i = \epsilon^{ijk} D_j A_k, \quad (19)$$

where $\mathcal{L}_n A_i$ is the Lie derivative of the spatial vector potential along n^μ (in coordinates this reduces to $\mathcal{L}_n A_i = (1/\alpha)(\partial_t A_i - \mathcal{L}_\beta A_i)$). In the simple case $\alpha = 1, \beta^i = 0$ this reduces to the familiar flat-space relations

$$E_i = -\partial_t A_i - D_i \Phi, \quad B^i = \epsilon^{ijk} D_j A_k. \quad (20)$$

Choosing a gauge simplifies these formulae: for instance the Lorenz gauge in curved space reads

$$\nabla_\mu A^\mu = 0 \quad \implies \quad -\mathcal{L}_n \Phi + D_i A^i + \Phi K = 0, \quad (21)$$

which when combined with the wave equation for A^μ yields decoupled hyperbolic equations for the potentials. In the Coulomb gauge ($D_i A^i = 0$) the scalar potential satisfies an elliptic constraint (Poisson-like) on each slice while A_i carries the radiative degrees of freedom.

3 General Relativity in the 3+1 Formalism

In this section we derive the 3 + 1 splitting of Einstein's equations and the associated geometric relations (Gauss, Codazzi, Mainardi, and Ricci). These relations link the four-dimensional curvature ${}^{(4)}R_{\mu\nu\rho\sigma}$ to the intrinsic geometry of a spatial hypersurface Σ_t and its extrinsic curvature.

We begin with a foliation of spacetime $(\mathcal{M}, g_{\mu\nu})$ by spacelike hypersurfaces Σ_t labelled by a time function t . The unit normal to the hypersurfaces is

$$n^\mu = -\alpha \nabla^\mu t, \quad n_\mu n^\mu = -1, \quad (22)$$

where α is the lapse function. The shift vector β^i defines the relation between the coordinate time vector ∂_t and the normal direction:

$$(\partial_t)^\mu = \alpha n^\mu + \beta^\mu, \quad \beta^\mu n_\mu = 0. \quad (23)$$

The spatial metric induced on Σ_t is

$$\gamma_{\mu\nu} = g_{\mu\nu} + n_\mu n_\nu, \quad (24)$$

which acts as a projection operator onto the slice: $\gamma^\mu{}_\nu = \delta^\mu{}_\nu + n^\mu n_\nu$. Its nonzero components in coordinates adapted to the foliation are γ_{ij} .

3.1 Extrinsic curvature and spatial covariant derivative

The extrinsic curvature of the slice is defined by the projected covariant derivative of the normal:

$$K_{\mu\nu} = -\gamma_\mu{}^\alpha \gamma_\nu{}^\beta \nabla_\alpha n_\beta, \quad (25)$$

with $K_{\mu\nu} n^\nu = 0$. On the slice this reduces to

$$K_{ij} = -\frac{1}{2} \mathcal{L}_n \gamma_{ij}. \quad (26)$$

The spatial covariant derivative associated with γ_{ij} is denoted D_i and satisfies

$$D_k \gamma_{ij} = 0. \quad (27)$$

The acceleration of Eulerian observers is

$$a_i = n^\mu \nabla_\mu n_i = D_i \ln \alpha. \quad (28)$$

3.2 Projections of the spacetime Riemann tensor

To obtain the 3+1 structure of Einstein's equations, we project the spacetime Riemann tensor ${}^{(4)}R_{\mu\nu\rho\sigma}$ using n^μ and $\gamma^\mu{}_\nu$. This yields the four fundamental relations (Gauss, Codazzi, Mainardi, Ricci).

Gauss Relation (all indices tangent to Σ_t)

Projecting all indices onto the slice gives

$${}^{(4)}R_{\alpha\beta\gamma\delta} \gamma^\alpha{}_i \gamma^\beta{}_j \gamma^\gamma{}_k \gamma^\delta{}_l = R_{ijkl} + K_{ik} K_{jl} - K_{il} K_{jk}, \quad (29)$$

where R_{ijkl} is the 3-dimensional Riemann tensor of (Σ_t, γ_{ij}) . Contracting twice yields the relation between four- and three-dimensional scalar curvatures:

$${}^{(4)}R = R + K^2 - K_{ij} K^{ij} - 2\nabla_\mu (n^\mu K - a^\mu). \quad (30)$$

Codazzi Relation (one normal, three tangent)

Projecting one index along n^μ gives

$${}^{(4)}R_{\alpha\beta\gamma\delta}\gamma^\alpha{}_i\gamma^\beta{}_j\gamma^\gamma{}_kn^\delta = D_jK_{ik} - D_iK_{jk}. \quad (31)$$

Mainardi Relation (two normal, two tangent)

The projection with two normals yields

$${}^{(4)}R_{\alpha\beta\gamma\delta}\gamma^\alpha{}_in^\beta\gamma^\gamma{}_jn^\delta = -\mathcal{L}_nK_{ij} + K_{ik}K^k{}_j + D_ia_j + a_ia_j. \quad (32)$$

3.3 Constraint equations from Einstein's equation

Einstein's equation,

$${}^{(4)}G_{\mu\nu} = 8\pi T_{\mu\nu}, \quad (33)$$

yields constraints when projected entirely along n^μ or once along n^μ and once tangent to the slice. Define the matter energy and momentum densities measured by Eulerian observers:

$$\rho = T_{\mu\nu}n^\mu n^\nu, \quad (34)$$

$$j_i = -T_{\mu\nu}n^\mu\gamma^\nu{}_i. \quad (35)$$

Hamiltonian Constraint

Projecting Einstein's equation twice along the normal gives

$${}^{(4)}G_{\mu\nu}n^\mu n^\nu = 8\pi\rho. \quad (36)$$

Using the Ricci relation (30) we obtain

$$R + K^2 - K_{ij}K^{ij} = 16\pi\rho. \quad (37)$$

This is the Hamiltonian (or scalar) constraint.

Momentum Constraint

Projecting Einstein's equation once along n^μ and once tangent gives

$${}^{(4)}G_{\mu\nu}n^\mu\gamma^\nu{}_i = 8\pi j_i. \quad (38)$$

Using the Codazzi relation (31) produces

$$D_j(K^j{}_i - \delta^j{}_i K) = 8\pi j_i. \quad (39)$$

Equations (37) and (39) are the constraint equations of general relativity within the 3 + 1 formalism.

3.4 Summary of the Four Fundamental Relations

$$\text{Gauss: } {}^{(4)}R_{ijkl} = R_{ijkl} + K_{ik}K_{jl} - K_{il}K_{jk},$$

$$\text{Codazzi: } {}^{(4)}R_{ijkn} = D_jK_{ik} - D_iK_{jk},$$

$$\text{Mainardi: } {}^{(4)}R_{ijn} = -\mathcal{L}_nK_{ij} + K_{ik}K^k{}_j + D_ia_j + a_ia_j,$$

$$\text{Ricci (contracted Gauss): } {}^{(4)}R = R + K^2 - K_{ij}K^{ij} - 2\nabla_\mu(n^\mu K - a^\mu).$$

These four identities form the geometric backbone of the ADM/York 3 + 1 decomposition and lead directly to the Hamiltonian and momentum constraints used in numerical relativity.

4 Numerical Methods for Maxwell's Equations

This section describes the numerical techniques used in the simulation and discusses the comparison between two possible methods of simulating, including initializing potentials and deriving fields from it and directly initializing fields. Numerical complexity is also discussed for both methods.

4.1 Spectral representation and use of the FFT

For periodic domains we adopt a Fourier spectral (pseudo-spectral) discretization. A spatial field $f(\mathbf{x})$ sampled on a regular N^3 grid is represented by its discrete Fourier coefficients

$$\tilde{f}(\mathbf{k}) = \mathcal{F}f(\mathbf{x}), \quad (40)$$

computed with the Fast Fourier Transform (FFT). Spatial derivatives are computed in spectral space as multiplications by $i\mathbf{k}$:

$$\mathcal{F}\partial_j f(\mathbf{k}) = ik_j \tilde{f}(\mathbf{k}). \quad (41)$$

The pseudo-spectral method evaluates nonlinear terms (or products) in physical space, using FFT $\leftrightarrow$ inverse FFT pairs and (if needed) dealiasing (e.g. 2/3-rule).

Key computational points:

- One forward or inverse 3D FFT on an N^3 grid costs $\mathcal{O}(N^3 \log N)$ operations.
- To compute all components of $\nabla \times$ or $\nabla \cdot$ spectrally requires multiplying each Fourier mode by $i\mathbf{k}$ and performing 3 inverse FFTs to return to physical space.
- Projection to a divergence-free field (Helmholtz projection) is performed in Fourier space by

$$\tilde{\mathbf{B}}^\perp(\mathbf{k}) = \tilde{\mathbf{B}}(\mathbf{k}) - \frac{\mathbf{k}(\mathbf{k} \cdot \tilde{\mathbf{B}}(\mathbf{k}))}{|\mathbf{k}|^2}, \quad \mathbf{k} \neq 0, \quad (42)$$

which enforces $\mathbf{k} \cdot \tilde{\mathbf{B}}^\perp = 0$ exactly for all retained modes.

4.2 Initializing via potentials vs directly initializing fields

Two common choices for producing initial data are:

Potentials first: specify vector potential $\mathbf{A}(\mathbf{x})$ and scalar potential $\Phi(\mathbf{x})$, enforce a gauge (e.g. Coulomb $\nabla \cdot \mathbf{A} = 0$), then compute

$$\mathbf{B} = \nabla \times \mathbf{A}, \quad \mathbf{E} = -\partial_t \mathbf{A} - \nabla \Phi. \quad (43)$$

Direct fields: specify $\mathbf{E}(\mathbf{x})$ and $\mathbf{B}(\mathbf{x})$ directly on the grid, projecting them onto the divergence-free subspace to reduce constraint violation.

Error characteristics

- When computing fields from smooth, divergence-free potentials using a spectral derivative, the resulting $\mathbf{B} = \nabla \times \mathbf{A}$ is divergence-free up to machine precision for all retained modes because spectral derivatives are exact for the represented Fourier basis. Thus initial divergence errors are negligible (limited only by roundoff and aliasing from truncation).
- Directly prescribing random or ad-hoc field components almost always produces significant $\nabla \cdot \mathbf{B} \neq 0$ unless a projection is applied. Even after projection, small-scale aliasing can leave residuals that the time integrator and nonlinear terms amplify.
- If initial data are required to satisfy constraints exactly (for convergence studies) the potentials approach is generally superior because a consistent gauge + spectral derivatives gives machine-accurate satisfaction of $\nabla \cdot \mathbf{B} = 0$ (and, if desired, constraints on $\mathbf{E}$ via Φ).

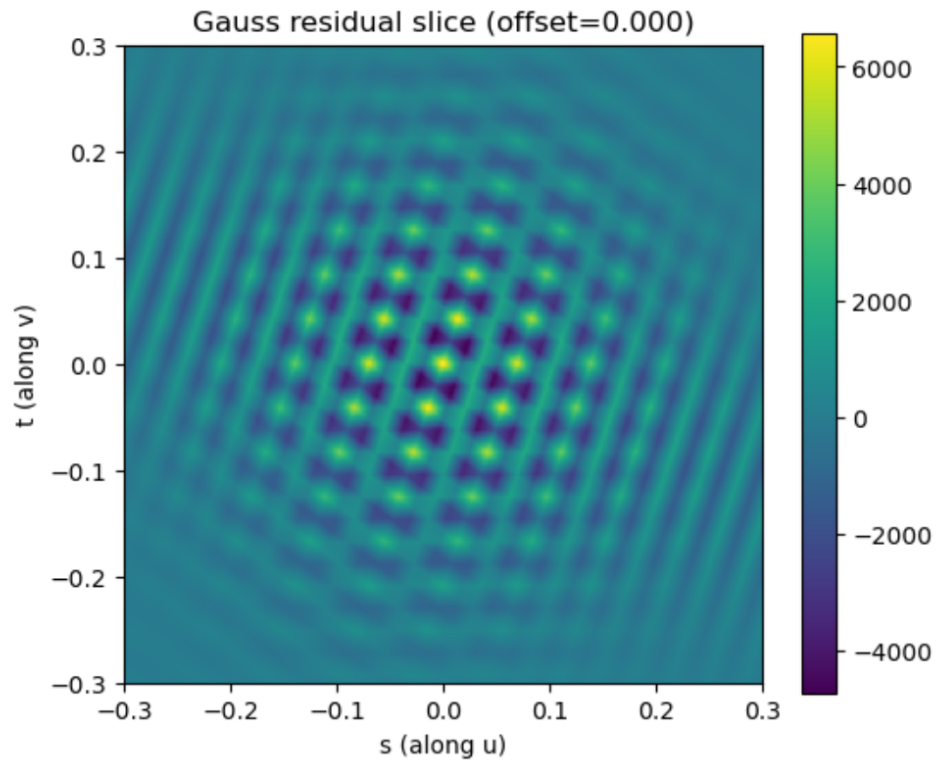


Figure 3: Residual Gauss error in Potential initialisation method

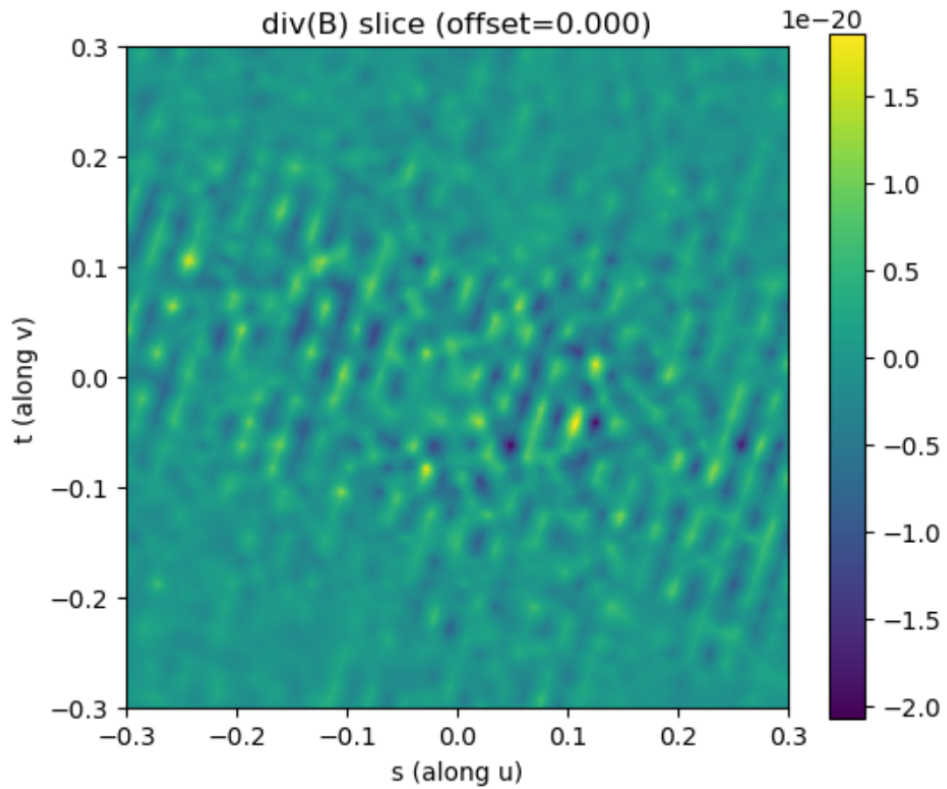


Figure 4: Residual div.(B) error in Potential initialisation method

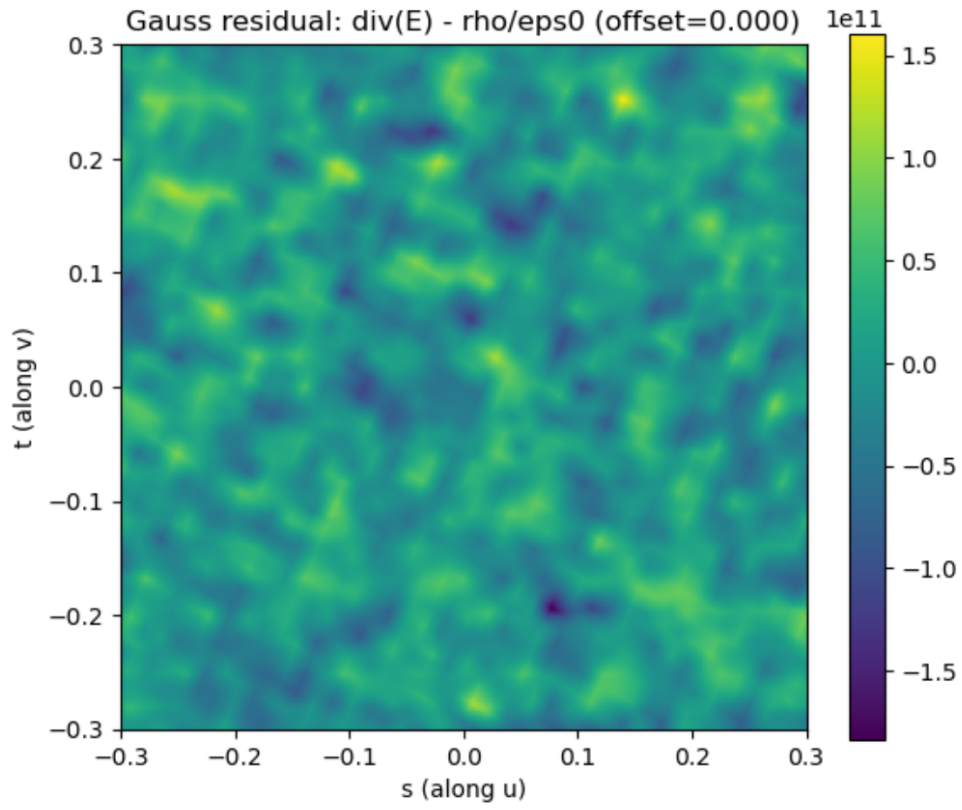
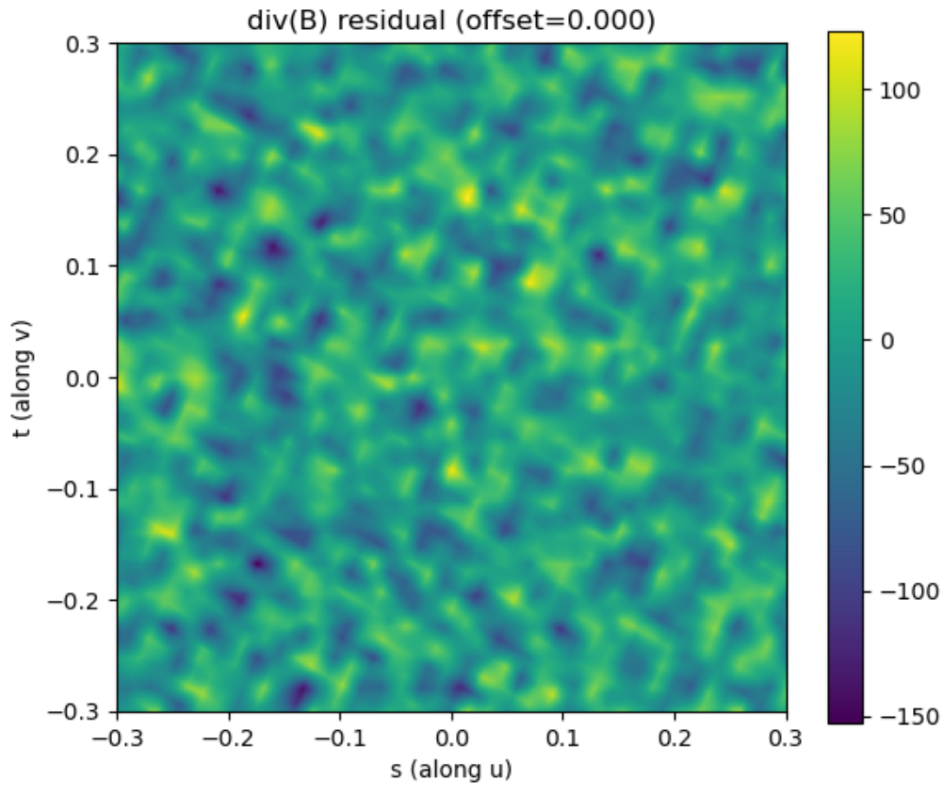


Figure 5: Residual Gauss error in Field initialisation method

Figure 6: Residual $\text{div}(\mathbf{B})$ error in Field initialisation method

Practical costs and trade-offs

- **Potentials method:** extra cost comes from storing and evolving potentials (additional fields), and converting between potentials and fields (spectral curls and gradients). Cost per conversion is dominated by FFTs ($\mathcal{O}(N^3 \log N)$). However, because this method produces low initial constraint violation, it often saves time otherwise spent in divergence-cleaning or small time-step reductions caused by nonphysical high-frequency modes.
- **Direct-field initialisation:** cheaper at setup (no curl computations needed at $t=0$) but potentially more expensive during the run because of required projection/cleaning operations and the possible need for increased resolution/time-filtering to control spurious modes. For random (high-frequency) initial fields the effective time-step or filtering cost can dominate.

4.3 Computational cost and practical timing

Theoretical scaling

Let N be the 1D grid resolution (total grid points $\propto N^3$). Denote by $T(N)$ the time per time-step. Then:

$$T_{\text{FFT-spectral}}(N) \propto n_{\text{FFT}} \cdot N^3 \log N, \quad T_{\text{FDTD}}(N) \propto C \cdot N^3, \quad (44)$$

so doubling $N \rightarrow 2N$ increases T roughly by a factor

$$\frac{T(2N)}{T(N)} \approx 2^3 \cdot \frac{\log(2N)}{\log N} \approx 8 \times (1 + o(1)). \quad (45)$$

In practice the logarithmic factor is small, so spectral methods scale approximately like $8\times$ work when doubling resolution in each dimension, while FDTD scales very nearly as $8\times$ but with a smaller per-point constant (no $\log N$ factor).

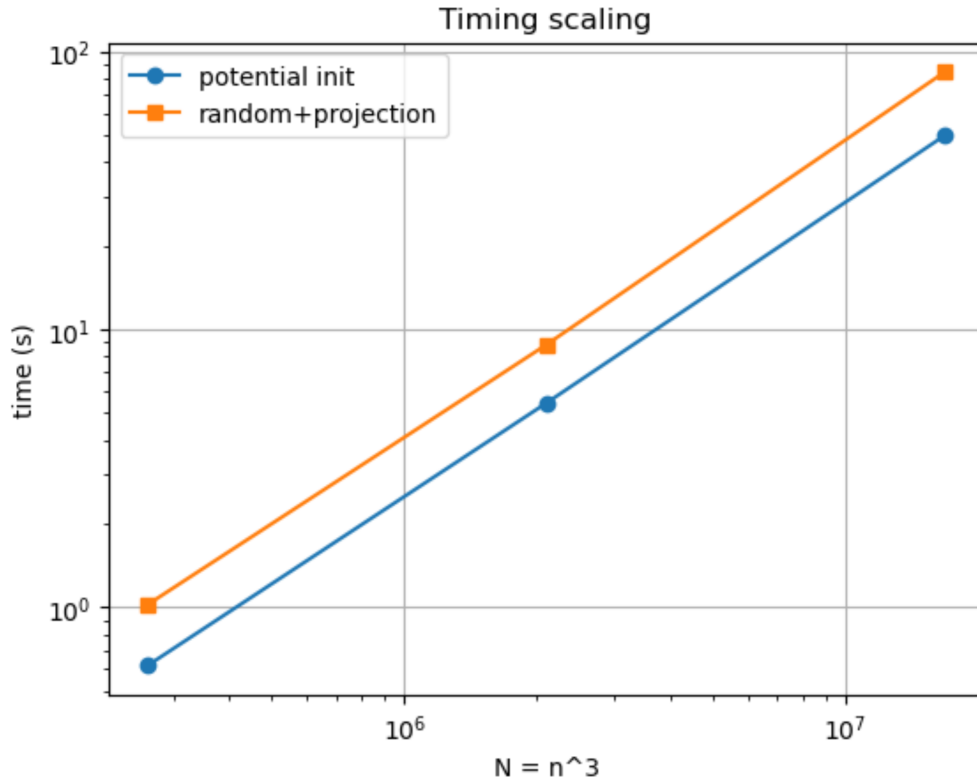


Figure 7: Plot of runtime(s) vs. number of grid points (N) obtained from simulation

Residuals obtained from the two methods of initialisation

n	N	pot $\ \nabla \cdot E\ $	pot $\ \nabla \cdot B\ $	proj $\ \nabla \cdot E\ $	proj $\ \nabla \cdot B\ $
64	262,144	7340.14	3×10^{-20}	7336.89	10.26
128	2,097,152	1902.99	9×10^{-20}	1898.96	15.48
256	16,777,216	480.04	2.1×10^{-19}	488.83	27.08

Table 1: Separated residuals for E and B under potential-based and projection-based initialisation.

4.4 Random initial fields vs updating-from-potentials

- **One-time initialisation cost:**

- Potentials: need spectral projections and curl computations (several FFTs) to produce $\mathbf{E}, \mathbf{B}$ at $t = 0$; cost is $\mathcal{O}(N^3 \log N)$ but paid only once (or each time you remake the initial state).
- Direct random fields: almost free to assign random numbers, but typically followed by a projection (FFT-based) or a divergence-cleaning solve to reduce constraint violation; this projection has the same $\mathcal{O}(N^3 \log N)$ cost.

- **Run-time cost:**

- If the potentials approach avoids repeated cleaning, it reduces the per-time-step overhead and often yields cleaner spectra that allow slightly larger time steps. This saving accumulates and for long runs can dominate the one-time setup cost.
- If direct initialisation leaves significant $\nabla \cdot \mathbf{B}$, you will pay per-step for cleaning (hyperbolic cleaning fields or frequent Helmholtz projections) or suffer accuracy/stability penalties that may force smaller time steps or more dissipative filters.

4.5 Boundary conditions and domain choices

- **Periodic domain:** best suited for Fourier spectral methods. All spectral transforms, derivatives, and Helmholtz projections are natural and inexpensive. Use this for homogeneous-wave and spectral-convergence tests.
- **Non-periodic domain:** require either a different spectral basis (Chebyshev) or hybrid approaches (spectral in some directions, finite-difference in others) and/or buffer layers. For wave-extraction we also use perfectly matched layers (PML) implemented in physical space; PMLs introduce additional fields and discrete operators and are usually combined with finite-difference stencils.
- **Yee-type staggered grids (FDTD):** advantage is exact discrete preservation of $\nabla \cdot \mathbf{B} = 0$ for appropriate updates; easier to implement PMLs and local boundary conditions.

4.6 Visualizing Dipole residuals using Potential Initialisation

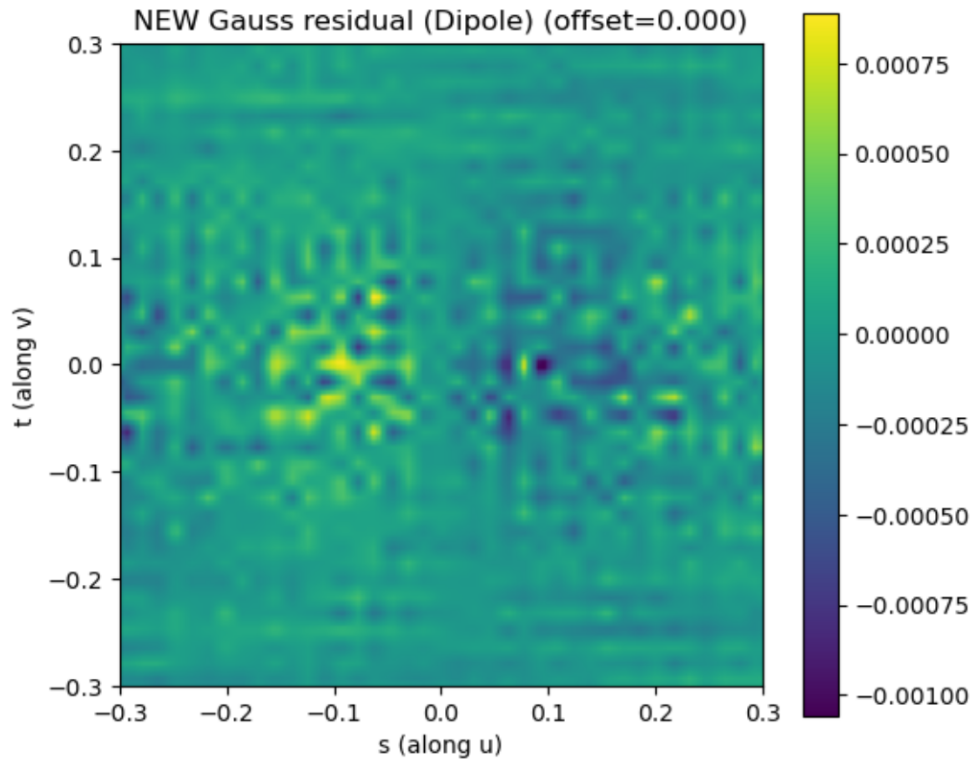


Figure 8: Residual Gauss error for electric dipole

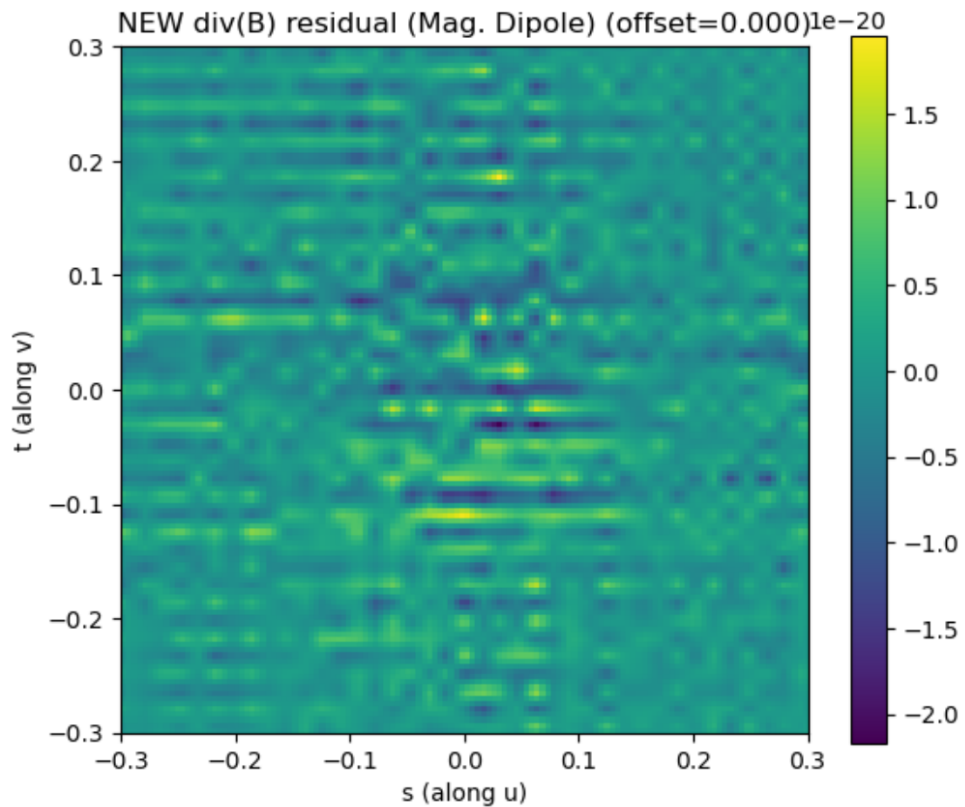


Figure 9: Residual div.(B) error for magnetic dipole

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